

Coffee Shop Sales Performance Analysis Using Microsoft Power BI

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1. Project Overview

This project is about a coffee shop sales company.

Main Business Scope: Retail Sales

The company sells:

- Coffee
- Tea
- Bakery products
- Drinking chocolate
- Coffee beans
- Other related products

The goal of this project:

- Track total orders
- Track revenue
- Track total cost
- Track net profit
- Check product and store performance

👉 The dashboard helps managers understand sales performance and make better decisions.

2. Dataset

Orders / Sales Transactions (Fact Table)

This is the main table. It contains all coffee shop transactions.

Each row = one transaction/order line

👉 This table shows what happened in sales.

It includes:

(Transaction ID, Transaction Date, Transaction Time, Transaction quantity, Product ID, Store ID)

Other details come from:

- Products Table: what we sell (Product ID, Product Category, Product Type, Product Detail, Price, Cost, Price)
- Store / Location Table: where sales happen (Store ID, Store Location, Latitude, Longitude)
- Date Table: when the sale happened (Year, Month, Day, Weekday)

3. Data Preparation

1. General Data Cleaning

This step is important to make sure the model works correctly all the time.

👉 We clean the data before using it.

Steps:

- Remove duplicates
- Remove empty rows
- Fix data types (Date, Time, Number, Text)
- Handle errors
- Check products and store names

👉 This makes the data clean, reliable, and stable.

2. Enhancements

In this step, we improve the data for better analysis.

👉 We add new columns and logic.

Added:

- Day column → to analyze data by day
- Hour / Time column → to analyze busy hours
- Day Order column → to sort weekdays from Monday to Sunday in the correct order

👉 This helps us:

- Analyze time better
- Track performance vs target

4. Data Modeling

We used a Star Schema model.

Tables:

- Fact Table → Orders / Sales Transactions
- Dimension Tables:
 - Products
 - Stores Location
 - Calendar

Relationships:

- Orders → Products (Product ID)
- Orders → Stores (Store ID)
- Orders → Calendar (Date)

👉 This helps:

- Fast dashboard performance
- Easy filtering
- Clear analysis by product, time, and location

5. Calculation (DAX)

We divide the analysis into 4 main parts:

1. Orders

👉 Goal: track number of orders.

KPIs:

- Total Orders: 👉 This shows how many orders we have.

Measure used:

- Total Orders = COUNTROWS (because each row is unique and represents one order)

2. Cost

👉 Goal: understand the total cost of sold products.

KPIs:

- Total Cost: 👉 This shows the total cost related to sold products.

Measure used:

- Total Cost = SUMX(Orders, Quantity × Cost)

3. Revenue and Profit

👉 Goal: understand revenue and profitability.

KPIs:

- Revenue: 👉 This shows total sales amount generated.
- Net Profit: 👉 This shows real profit after deducting cost.
- Margin Profit: 👉 This shows profit percentage.

Measures used:

- SUMX + RELATED (because Price and Cost are in another table “Products”)

👉 We calculate:

Sales = Qty × Price

Cost = Qty × Cost

👉 This is better than SUM because data is split per row

4. Performance Analysis

👉 Goal: compare performance by product, time, and store location.

KPIs and visuals:

- Orders by store location
- Orders by product category
- Orders by product type
- Revenue by product category
- Profit by product category
- Orders by time and weekday

Time Intelligence

👉 Goal: track change over time.

KPIs:

- Month-over-Month change: 👉 This shows growth compared to the previous month.

Measures used:

- Use Date table for monthly trend analysis
- Use CALCULATE to compare current period with previous period
- Use icons (▲ / ▼) to show trend direction

👉 Special behavior:

- Show monthly trend for orders and revenue
- Highlight increase or decrease compared to last month

6. Dashboard Layout

👉 Goal: make dashboard clear and easy to use.

The dashboard is divided into 3 main analysis pages:

(Overview – Financial – Products)

👉 Each page focuses on one business area.

General Flow

User reads each page like this:

- KPI (What happened?)
- Trend (How it changed over time?)
- Drivers (Why it happened?)

Each page analyzes the KPI main drivers as following:

1. Overview Page

This page gives a high-level summary of coffee shop sales performance.

- Total Orders → overall transaction volume
- Total Cost → total cost of sold products
- Revenue → total sales amount
- Net Profit → real profit
- Margin Profit → profit percentage
- Orders by Month → monthly order trend
- Orders by Time → busy hours
- Orders by Store Location → branch comparison
- Orders by Product Category → category demand

- Orders by Product Type → detailed product demand

2. Financial Page

👉 In this page, financial analysis focuses on Revenue, Total Cost, Net Profit, and Margin Profit.

👉 This helps the user understand financial performance from different points of view.

- Revenue by Month → financial trend over time
- Revenue by Time → high revenue hours
- Profit Waterfall → relationship between revenue, cost, and net profit
- Revenue by Product Category → strongest financial categories
- Financial Table → orders, revenue, and net profit by product category

3. Products Page

This page focuses on detailed product performance and time-based behavior.

- Orders by Time and Weekday → daily and hourly demand pattern
- Product Category by Month → category performance over months
- Product Type by Month → detailed product performance over months
- Top products → products with the highest demand
- Bottom products → products with the lowest demand

👉 This structure helps users understand:

- Where the strongest sales are
- Which products are driving revenue
- Which store locations perform better
- Which hours and days have the highest demand
- What is causing the change in performance

7. Interaction

👉 Goal: give the user flexibility to explore data.

Added features:

- Slicers → allow the user to filter data easily (Month, Day, Store Location, Products)

👉 This helps the user focus on specific data.

- Navigation Buttons → allow the user to move between dashboard pages (Home, Overview, Financial, Products)

👉 This makes the report easy to use.

- Measures Metric Parameter → allows switching between selected measures (Revenue / Net Profit depending on the visual setup)

👉 This gives dynamic analysis in one visual.

- Top 10 / Bottom 10 Buttons → allow product ranking analysis

👉 This helps identify best-selling and low-selling products.

👉 These interactions make the dashboard dynamic and user-friendly

8. Dashboard KPIs Summary

KPI	Value	Meaning
Total Orders	149K	Total number of orders / transactions
Total Cost	\$378.1K	Total cost of sold products
Revenue	\$700.8K	Total sales revenue
Net Profit	\$322.7K	Revenue after deducting cost
Margin Profit	46%	Profit percentage compared to revenue

9. Dashboard Pages Summary

Page	Purpose	Main Visuals
Orders	Provides a high-level summary of sales performance	KPI cards, orders trend, orders by time, orders by location, product category, product type
Financial	Analyzes revenue, cost, profit, and margin performance	KPI cards, revenue trend, revenue by time, profit waterfall, financial table
Products	Provides detailed product and time analysis	Matrix tables, product category by month, product type by month, top and bottom products

10. Conclusion

👉 Goal: support daily decisions and business monitoring.

This dashboard helps managers to:

- Monitor daily sales and orders
- Understand if business is running well or not
- Track revenue, cost, net profit, and margin profit
- Identify best-selling and low-selling products
- Compare store locations performance
- Find peak sales hours and days
- Take fast decisions

👉 Example:

- Increase focus on high-selling products
- Improve low-performing products
- Support branches with weak performance
- Plan staff based on busy hours
- Improve profitability by controlling cost

👉 In simple words:

This dashboard turns raw coffee shop sales data into clear business insights and actions.

👉 Result:

Better decisions, faster response, higher profitability, and better business growth